

ZeCheng (Danny) Deng

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Summary

Software engineer with a Master of Cybersecurity and two development co-op terms. Builds across three areas: full-stack web with React and Next.js, offensive and cloud security tooling, and gameplay programming in Unity and C++. CompTIA Security+ and ISC2 CC certified.

Technical Skills

Certifications: CompTIA Security+ (2026), ISC2 Certified in Cybersecurity, CC (2026)

Languages: Python, JavaScript, TypeScript, C++, C#, Java, C, Go, SQL, PHP

Frontend: React, Next.js, Redux, Tailwind CSS, HTML5, CSS3, Ant Design, Recharts

Backend & Data: Node.js, Express, REST APIs, GraphQL, PostgreSQL, MongoDB, Redis, DynamoDB

Cloud & DevOps: AWS (EC2, S3, VPC, IAM, Lambda, API Gateway, Cognito), Terraform, Docker, nginx, Vercel, Git

Security: Wireshark, tcpdump, Scapy, nmap, gobuster, OWASP ZAP, GDB, iptables, Kali Linux

Game & Graphics: Unity, OpenGL, GLSL, ECS architecture

AI & LLM: LLM agent pipelines, MCP, retrieval-augmented generation (BM25), Ollama, Claude Code

Experience

JavaScript Developer, Co-op | Semios, Vancouver, BC

Jan – Aug 2022

- Built and maintained React components for an agricultural data platform, delivering 3 to 5 tickets per sprint in an Agile workflow.
- Built reusable UI components with the Ant Design library, including data tables, forms, and interactive map interfaces.
- Implemented form validation, content sorting, and date and time handling using moment.js.
- Resolved frontend defects and shipped UI/UX improvements in close collaboration with the QA team.

Web Developer, Co-op | Bluevec Technologies, Vancouver, BC

Sep – Dec 2021

- Built the company's 7-page marketing site in React from business-owner requirements, authoring the majority of commits on the project.
- Deployed the static build to an AWS EC2 instance running nginx, and served the homepage background video from an S3 bucket.
- Wrote Go unit tests covering two features to improve reliability of the internal tooling.
- Built an image management utility with GoCV handling timestamp processing and automated file cleanup.

Teaching Assistant | University of British Columbia

May – Aug 2022 & 2023

- Supported CPSC 455 Applied Industry Practices, a project course on the MERN stack, running 4 office-hour hours per week.
- Graded 5 students' assignments per week in person, giving one-on-one feedback on code structure and debugging.

Projects

APEX, Autonomous Penetration Testing Framework | Python, LLM Agents, MCP

Jan – May 2026

- Rewrote an inherited v1 codebase into a coordinator-driven v2 pipeline, evaluated on 42 HackTheBox machines with a 71% success rate (30/42) across Linux and Windows targets.
- Built a multi-agent pipeline in which recon agents (nmap, gobuster, OWASP ZAP) and execution agents dispatch LLM-generated bash scripts to Kali Linux over a custom MCP-to-HTTP REST bridge.
- Designed a three-class failure classifier (FIXABLE / FUNDAMENTAL / SCRIPT_ERROR) with three independent loop-detection counters that auto-fail stalled tasks instead of retrying forever.
- Implemented dual BM25 retrieval pipelines, one over offensive payloads and one over historical failures, to guide exploit selection and avoid repeating dead ends.

Personal Website (dannydeng.dev) | *Next.js, React, PostgreSQL, Redis*

2025 – Present

- Designed, built, and deployed a Next.js App Router site on Vercel, owning the frontend, data layer, and deployment end to end.
- Aggregated 6 Last.fm and iTunes endpoints in parallel with Promise.all, normalizing differing response shapes behind a single internal API.
- Reduced upstream API calls using an Upstash Redis cache on a 1-hour TTL, refreshed by a daily Vercel cron job.
- Built semantic photo search from CLIP embeddings stored in Supabase PostgreSQL, queried by pgvector cosine similarity.

Threat Intelligence Dashboard (Team Project) | *Next.js, TypeScript, n8n, PostgreSQL*

Sep – Dec 2025

- Built scheduled ingestion workflows in n8n for Hacker News and phishing feeds and co-authored the CVE feed, each pulling from an external API, transforming records in a JavaScript step, and inserting them into PostgreSQL.
- Built the dashboard frontend in Next.js and TypeScript: a sortable table over the ingested records, three Recharts views covering CVEs over time, phishing trends, and trending topics, and severity, source, and time-range filters.

Network & Binary Security Labs (SFU) | *Scapy, Wireshark, GDB*

Jan – Apr 2025

- Executed TCP SYN flood, RST injection, and Telnet session hijacking against a 3-node Vagrant network using Scapy-crafted packets, then verified mitigation through SYN cookies.
- Performed DNS spoofing attacks, analyzed the resulting traffic in Wireshark and tcpdump, and wrote iptables firewall rules to block them.
- Exploited buffer overflow, format string, return-to-libc, and return-oriented programming chains on x64 Linux binaries with NX enabled, locating gadgets and building payloads in GDB.

Cloud Infrastructure & IAM Labs (SFU) | *AWS, Terraform*

Sep – Dec 2024

- Deployed a segmented AWS VPC with public and private subnets, Internet and NAT gateways, route tables, and security groups, hosting a Node.js server on EC2 backed by DynamoDB.
- Reproduced the same stack as Terraform infrastructure as code with repeatable deploy and destroy cycles.
- Built a serverless REST API on Lambda and API Gateway, securing the POST endpoint with Cognito JWT authorization and validating access control in Postman.
- Audited IAM users, groups, and roles against CIS benchmarks, checking MFA coverage, key rotation, and least privilege.

2D-Plat | *Unity, C#*

May 2026 – Present

- Building a 2D action platformer around the Command design pattern, using an ICommand interface and an InputHandler dispatcher to decouple input from behaviour.
- Implemented a WaveManager enemy spawning system, enemy behaviour, camera follow, and a player health bar.

Eggly Buddy | *C++, OpenGL, ECS Framework*

Sep – Dec 2023

- Programmed a shell-game mini-game and in-game pet mechanics in C++ on a custom entity component system framework.
- Wrote custom GLSL shaders driving a dynamic parallax background and sprite animation.
- Built a save and load system with the nlohmann/json library for persistent game state.

Swag the North | *Unity, C#*

Sep 2023

- Placed top 12 out of 200+ projects at Hack the North with a team of four.
- Built core gameplay systems including player control, item spawning, entity health, UI, and game state management, and owned the team's Git and pull request workflow.

Education

Simon Fraser University | *Master of Cybersecurity*

Sep 2024 – May 2026

University of British Columbia | *BSc in Computer Science, with Distinction*

Sep 2018 – May 2024

- Relevant coursework: Video Game Programming, Computer Graphics, Distributed Systems, Applied Industry Practices.